Physics Week 9 Model Benchmark v1

DeepSeek API vs Codex CLI headless on the text-only held-out split

Best completed condition
Codex CLI + C2
candidate-v2, held-out test

Held-out exact agreement
89.8%
Codex C2 vs gold scores

Held-out total MAE
1.08
lower is better

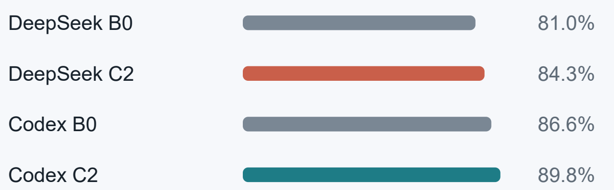
Executive readout. This v1 report compares the two completed physics Week 9 text-only model families: DeepSeek public API and Codex CLI headless. Both providers were run with the same baseline packet and candidate-v2 packet on the development split and on the held-out test split. The strongest completed result is **Codex CLI + candidate-v2** on held-out test: exact agreement 0.898, total-score MAE 1.083, within-1-point rate 0.667, and severe-error rate 0.333.

0.1. What Is Being Compared

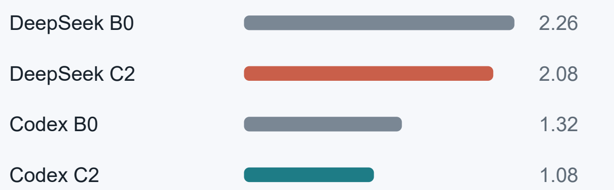
Condition	Provider / model	Input	Purpose
B0	DeepSeek v4-pro or Codex gpt-5.5	text-only transcript	Original baseline grading packet.
C2	same provider / model	text-only transcript	Candidate-v2 grading skill packet: more explicit scoring rules, stricter schema, and process-aware handling for calculation answers.
Dev split	8 students	96 subquestion rows	Used for debugging/calibration. Not the final evidence split.
Held-out split	18 students	216 subquestion rows	Primary comparison split. It was kept separate from prompt tuning.

0.2. Held-out Test: Main Evidence

Exact agreement, higher is better



Total-score MAE, lower is better



Held-out condition	Exact	SubQ MAE	Total MAE	Within 1 pt	Severe err.
DeepSeek B0	0.810	0.200	2.264	0.333	0.444
DeepSeek C2	0.843	0.185	2.083	0.500	0.444
Codex B0	0.866	0.126	1.319	0.667	0.333
Codex C2	**0.898**	**0.100**	**1.083**	**0.667**	**0.333**

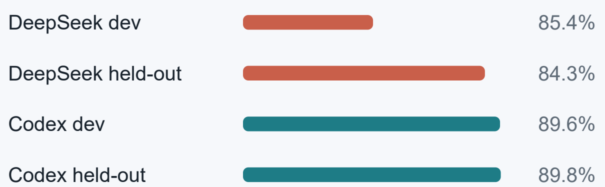
Interpretation. Candidate-v2 improves exact agreement and MAE for both completed providers. Codex CLI is stronger than DeepSeek on every held-out metric in this v1 comparison. However, severe-error rate remains non-trivial, so this is a benchmark result rather than a production-ready grading claim.

0.3. Baseline vs Candidate-v2 Deltas

Positive exact-agreement deltas mean candidate-v2 matched the gold score more often. Negative MAE deltas mean candidate-v2 reduced scoring error.

Held-out delta	DeepSeek C2-B0	Codex C2-B0	Readout
Exact agreement	+0.032	+0.032	Both providers gain the same absolute exact-agreement lift on held-out test.
Subquestion MAE	−0.015	−0.027	Codex shows a larger reduction in subquestion-level error.
Total-score MAE	−0.181	−0.236	Both improve; Codex reaches the lower final error.
Within-1-point rate	+0.167	+0.000	DeepSeek improves proximity; Codex was already higher and stays stable.
Severe-error rate	+0.000	+0.000	No held-out improvement. This is the main remaining weakness.

Candidate-v2 exact agreement



Candidate-v2 severe-error rate

0.4. Development Split Sanity Check

The development split is useful for detecting obvious packet failures, schema failures, or candidate regressions before spending more runs on held-out data. It should not be used as the final conclusion because the prompt packet was calibrated using development feedback.

Dev condition	Exact	SubQ MAE	Total MAE	Within 1 pt	Severe err.
DeepSeek B0	0.792	0.221	2.156	0.375	0.375
DeepSeek C2	0.854	0.104	0.750	0.750	0.125
Codex B0	0.875	0.099	0.438	0.875	0.125
Codex C2	0.896	0.081	0.594	0.875	0.125

Development caveat. Codex candidate-v2 improves exact agreement on dev, but its dev total-score MAE is slightly worse than Codex baseline. Held-out test reverses that concern: Codex candidate-v2 lowers total-score MAE from 1.319 to 1.083. This is why the held-out split is the primary evidence.

0.5. Good-enough Gate

Gate	Status	Evidence
All expected students pass schema validation	Pass	DeepSeek and Codex held-out runs passed validation.
Held-out exact agreement at least 0.84	Pass	DeepSeek C2 = 0.843; Codex C2 = 0.898.
Held-out exact agreement improves over baseline	Pass	Both providers improve by +0.032.
Held-out total-score MAE at most 2.10 and improves	Pass	DeepSeek C2 = 2.083; Codex C2 = 1.083.
Held-out within-1-point rate at least 0.50	Pass	DeepSeek C2 = 0.500; Codex C2 = 0.667.
Severe-error rate not worsened	Pass with caveat	No worsening, but rates remain high: DeepSeek 0.444, Codex 0.333.

0.6. Reproducibility Notes

Code version	Report summary: 780e75a. DeepSeek held-out: 9cce18378abb19d817040cb56599457108d7d575. Codex held-out: 780e75a.
Data policy	Raw Data/ and model outputs are private/ignored. This report contains only aggregate metrics and reproducibility pointers.
DeepSeek source	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/held-out-metrics-strict-schema.json
Codex source	Data/physics/benchmark/runs/physics-week9-codex-headless/codex-heldout-G1-baseline-vs-candidate.metrics.json
Codex engine	Codex CLI 0.133.0, headless mode, model label gpt-5.5.
Current missing runs	Kimi and Claude are not included in this v1 report. They should be added in v2 after mentor-side runs return complete outputs.

0.7. Conclusion for v1

For the completed physics Week 9 text-only benchmark, **Codex CLI + candidate-v2** is the current best-supported condition. It clears the pre-set good-enough bar on held-out exact agreement, total-score MAE, and within-1-point rate, and it outperforms DeepSeek candidate-v2 on the held-out split.

This is still not the final model-selection report. The next report version should add Kimi and Claude Code runs, include the same held-out metrics, and keep severe-error analysis as a first-class criterion rather than treating exact agreement alone as sufficient.